Explicit counterexamples to the Jacobian and Hessian conjectures, and weak Markus–Yamabe fields, decided in exact rational arithmetic

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Abstract

In July and August 2026 the Jacobian conjecture of Keller (1939) was refuted in dimension three by an explicit polynomial map, an infinite family and a geometric explanation followed within days, a five-variable counterexample to the Hessian conjecture was derived from it, five further Keller counterexamples in dimensions three to five were published, and the weak Markus–Yamabe conjecture was refuted in dimension 14 by a vector field built on the same map. Every one of these objects is finite data, and every claim made about it reduces to finitely many identities over the rationals. We re-decide them all in exact rational arithmetic, with no code shared with any of the authors: the Jacobian or Hessian determinant is expanded symbolically and compared with the stated constant coefficient by coefficient, every published witness is re-evaluated as a fraction, and for the vector fields the nilpotency of $JX + I$ is decided as a polynomial identity. The first record holds 11 certificates, 8 re-decided from sources pinned by hash and 3 built here from new curves through the published tangent-sweep mechanism; a verifier of one file in the Python standard library re-derives all 11, and the published witnesses are then discarded and the collisions found again blind, as pairwise disjoint Krawczyk boxes on the exact map (7 of 9 cells; the cells that return a single box or none, 1 and 1 respectively, are reported as such). The second record decides 8 further claims: Gao's maps G, F_4, F_5, F_6 and the two Markus–Yamabe fields in dimensions 14 and 18 hold whole; F_7 is decided in part, its non-injectivity and the determinants of its factors, the whole determinant resting on the paper's factorisation lemma because its components have up to 25,518 terms. What is not decided — generic fiber sizes, coordinate equivalence of the generated maps, anything in the plane — is stated as such, and the moduli question the generator raises is left open.

1 Introduction

A polynomial map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ with a polynomial inverse has, by the chain rule, a Jacobian determinant that is a nonzero constant. Keller [1] asked whether the converse holds: whether every *Keller map*, a polynomial map with $\det J(F) \in \mathbb{C}^\times$, is a polynomial automorphism. The

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question became widely known through Bass, Connell and Wright [2] and van den Essen's book [3]; Wang [4] proved it for maps of degree at most two in every dimension, and Moh [5] for maps of degree at most 100 in the plane. Two structural facts frame any counterexample. By Białynicki-Birula and Rosenlicht [6] an injective polynomial map $\mathbb{C}^n \to \mathbb{C}^n$ is bijective with polynomial inverse, so a Keller counterexample must fail injectivity; and since a Keller map is étale, distinct preimages cannot merge over a target point, so injectivity can only fail through preimages escaping to infinity.

On 19 July 2026 Alpöge announced an explicit polynomial map $\mathbb{C}^3 \to \mathbb{C}^3$ with $\det J(F) \equiv -2$ that identifies three distinct rational points [8]. Gallagher [9] gave, the next day, a one-variable construction producing for every $d \geq 3$ a dimension-three Keller counterexample of generic fiber degree d; Tao [10] and Speyer [11] wrote expositions, Speyer isolating the mechanism as a *tangent sweep* of a plane curve conjugated by monomial maps. Meng and Yang [12] derived from the map, through a six-variable doubling and a one-variable Schur descent, an integer polynomial in five variables of total degree 14 with constant Hessian determinant 128 whose gradient is not injective, which refutes the Hessian conjecture HC_5. Gao [13] generalised the sweep from plane curves to direction fields on hypersurfaces and worked it out in five new explicit maps, one in dimension three of geometric degree 4 and two each in dimensions four and five, of geometric degrees 5, 10, 6 and 12; the paper's title footnote records that "AI disclosure: The main idea and framework are due to the author, and Claude Fable 5 assisted in the proofs and in the writing up of the paper.". Castañeda, Honorato and Valenzuela-Henríquez [14] turned the map into dynamics: a chain realisation theorem that converts any polynomial Keller map $F = I + H$ of $\mathbb{R}^n$ into a polynomial vector field on $\mathbb{R}^N$, $N = \sum_i \max(d_i, 2) - n$, whose Jacobian has spectrum $\{-1\}$ at every point and whose zeros are in bijection with a prescribed fiber of F. Applied to the announced map this gives a field of degree 7 on $\mathbb{R}^{14}$ with three zeros, so the weak Markus–Yamabe conjecture (every C^1 Hurwitz map of $\mathbb{R}^n$ is injective, the weak form of [15]; the classical form holds in the plane [16, 17] and fails from dimension three by the polynomial example of [18]) fails in dimension 14 and above; a second, independent reduction gives a cubic field on $\mathbb{R}^{18}$.

All of this is finite data: polynomials with rational coefficients, rational points, stated constants. Each paper reports that it checked its identities in exact arithmetic, and one announcement has since been formally verified in Isabelle/HOL, as cited in [14] [19]. What none of that supplies is a decision made with no code and no arithmetic in common with the claimant, recorded in a form a stranger can re-run, and able to say *refused* rather than *yes* when a claim outruns what the record holds. That is what this paper reports. The contribution is not the arithmetic, which is elementary, but the decision grammar, verdict plus witness plus refusal as a result, and the public, re-runnable record behind every number printed here: each constant in this document is a macro written by a program from the certificate files, and the program refuses to write the document when a record no longer says what a sentence needs.

What is decided, in one paragraph. Record I (§§4–5) holds 11 certificates: the announced map, in dimension 3 and once more with identity coordinates adjoined at $n = 8$; Meng–Yang's Hessian polynomial; Gallagher's seeds $d = 2, \ldots, 5$ and the paper's "genuinely distinct" member, each rebuilt from the seed under the paper's own gauge; and 3 maps generated here from curves of our own choosing through the published mechanism, which match no printed map. Every certificate is decided twice, by this repository's instrument and by a standard-library Python verifier that shares no arithmetic with it, and the 8 published ones are decided against bytes pinned by SHA-256. The published witnesses are then discarded and the collisions found again

blind. Record II (§§6–7) decides Gao's five maps and the two Markus–Yamabe fields together with the cubic map on eleven variables behind the second field: 7 of 8 claims hold whole, one holds in part. Section 8 states the moduli question the generator raises, open; §9 lists what is not claimed.

A word on the word. Throughout, a statement is *certified* when it has been decided true by a finite exact computation whose inputs, steps and result are recorded in a form a third party can re-run from the record alone; CERTIFIED, PARTIAL and REFUTED are the verdict words the ledgers use. Elsewhere we say *decided* or *proved*. No floating-point number participates in any verdict below; floats appear only in a screen that may prune and never admit, and in the hunter of §5, whose every candidate is then certified exactly.

2 Three claim shapes, and what decides each

Lemma 1 (Keller counterexample)**.** *Let $F\colon \mathbb{C}^n \to \mathbb{C}^n$ be a polynomial map with $\det J(F) \equiv c \in \mathbb{Q}^\times$ as a polynomial identity, and let $p \neq q$ be points with $F(p) = F(q)$. Then F is a Keller map that is not a polynomial automorphism, so the Jacobian conjecture is false in dimension n, and, adjoining identity coordinates, in every dimension $m \geq n$.*

Proof. An automorphism is injective. For $m > n$ the map $(F, x_{n+1}, \ldots, x_m)$ has the same determinant and the padded points still collide. □

Lemma 2 (Hessian counterexample)**.** *Let $\Psi \in \mathbb{Q}[x_1, \ldots, x_n]$ have $\det \operatorname{Hess} \Psi \equiv c \in \mathbb{Q}^\times$ and let $\nabla\Psi(p) = \nabla\Psi(q)$ with $p \neq q$. Then $\nabla\Psi$ is a non-injective Keller map, so the Hessian conjecture HC_n, in the form that the gradient of a polynomial with nonzero constant Hessian determinant is a polynomial automorphism, is false in n variables.*

Proof. $J(\nabla\Psi) = \operatorname{Hess}\Psi$; apply Lemma 1. (The reduction of the Jacobian conjecture to such symmetric maps is [7]; Meng–Yang state the consequence as the formal Legendre transform of Ψ failing to be a polynomial.) □

Lemma 3 (weak Markus–Yamabe counterexample)**.** *Let $X\colon \mathbb{R}^N \to \mathbb{R}^N$ be a polynomial vector field such that $(JX + I)^k = 0$ holds as an identity of matrices of polynomials for some k, and let X vanish at two distinct points. Then every eigenvalue of $JX(Z)$ equals -1 for every $Z \in \mathbb{R}^N$ (indeed for every $Z \in \mathbb{C}^N$), so X is a Hurwitz map that is not injective, and the weak Markus–Yamabe conjecture is false in dimension N and, by $X_n(Z, \xi) = (X(Z), -\xi)$, in every dimension $n \geq N$.*

Proof. At each point $M = JX(Z) + I$ is a nilpotent matrix, whose only eigenvalue is 0; the eigenvalues of $JX(Z) = M - I$ are therefore all -1. A map with two zeros is not injective. The block-diagonal extension keeps the spectrum and the zeros. □

Each lemma turns a conjecture's failure into finitely many facts about explicit rationals: a polynomial identity (a determinant, or a matrix power, with every nonconstant monomial required to cancel), exact evaluations at listed points, and pairwise distinctness. That is the entire content of a certificate, and it is why the verdicts below are decisions rather than samples. The hypotheses are also exactly where a claim can fail: a determinant that is not constant refutes nothing and is reported REFUTED, as is a witness that is off by 10^{-9} or a point listed twice.

3 The instrument

Arithmetic. Polynomials are sparse maps from exponent vectors to exact rationals, coefficients as arbitrary-precision integers (JavaScript `BigInt` fractions in the instrument, `fractions.Fraction` in the verifier and in the second record's decider). Partial derivatives are formal. The determinant of an $n \times n$ matrix of polynomials is expanded symbolically: the instrument and the verifier use a cofactor recursion over unused columns with zero pruning, the second record's decider a Laplace expansion organised as dynamic programming over column subsets. In every case the result is a polynomial, and the verdict is read off it: the constant it must equal and the absence of every other monomial. For the vector fields, $(JX + I)^k$ is multiplied out until it vanishes or k exceeds N.

Witnesses. Each listed point is substituted into each component in exact arithmetic and compared with the claimed image coordinate by coordinate as fractions; the points are compared pairwise. A float screen samples the determinant at one point before the symbolic expansion is attempted; it can only prune, and nothing about a passing sample is believed.

Controls. A verifier that has never rejected a plausible forgery has an unknown false-accept rate, so the instrument's gate plants forgeries that must be refuted before a verdict is accepted: one coefficient of the announced map moved by 10^{-6} (the determinant must stop being constant), one collision point moved by 10^{-9}, one point listed twice, a curve whose twist equations are violated by 10^{-6} (the generator must refuse to emit it), a seed violating the published gauge, a forged source hash, a source with no recorded pin, a perturbation of the Hessian polynomial, and the fiber counter run on a map that provably has one preimage. At the run that wrote this document the gate passed 32 checks with 9 such red controls fired. The second record's battery passed 19 checks with 6 of 6 red controls fired (a coefficient of G moved, a term of the 14-dimensional field moved to the next component, a witness moved by 1/1000, a printed zero moved, a coefficient of Φ moved).

Independence. The eleven certificates of Record I are detached from the machine that produced them into one file, `certs/keller-certificate.json` (SHA-256 prefix `00a0b36d4720`, exported at git `be82e12`): every polynomial as an explicit list of monomials with exact rational coefficients, every witness as exact rationals, the claimed constant, and the hash of every source the entry was transcribed from. A verifier of one file, `tools/verify_keller.py`, in the Python standard library only, re-derives the Jacobian by formal differentiation, expands the determinant, evaluates the witnesses, re-hashes the pinned sources, and then forges a coefficient and requires its own audit to fail. It decided 11 entries with 0 failures, re-hashed the 97 files of the certificate's pin table, and its red control fired. The record of the second ledger is checked by re-derivation: `tools/run-polymaps-ledger.py --check` recomputes every row from the pinned TeX and refuses on any difference, and the page that reports it will not build otherwise.

Provenance. The announced map arrived as a post with no canonical bytes of its own. Meng–Yang's preprint prints the map and all three witnesses in full, so the two announced-map rows pin that file (`mengyang2026.pdf`, SHA-256 prefix `090c1735997c`), which is also the Hessian entry's source; the Gallagher rows pin `gallagher2026.pdf` (`1782eefa569a`). The second record mirrors the TeX of both papers under their CC BY 4.0 licence (prefixes `c644040df103` and

`a5cf5acfa67a`; the PDFs pinned at `483208235e32` and `5a7b6ecf19a0`) and extracts every map with a parser that refuses any text it cannot consume. A drifted source refuses the certificate rather than certifying against a memory of it.

Lanes. Every certificate of Record I belongs to exactly one of three lanes, computed from the record rather than asserted: *transcribed* (a published map with its witnesses, bytes held), *rebuilt from a seed* (the paper published a recipe and a seed; the map is regenerated here with every gauge condition verified, then decided) and *generated* (a curve chosen here, pushed through the published mechanism). The counts are 3, 5 and 3. The distinction is the most useful fact on the page, and Table 1 draws it on every row.

The generator. The tangent sweep, in Gao's account [13] of Speyer's mechanism, is implemented as an exact-rational recipe. A plane curve $K(w) = (p(w), q(w))$ with $q'(w) = \frac{w}{2}p'(w)$ has sweep $S(\gamma, w) = (p(w) + 2\gamma,\ q(w) + \gamma w)$ with $\det J(S) = 2\gamma$. The monomial twist $\gamma = g_0 + a\,xy + b\,x^2 z$, $u = 1 + xy$, $w = \gamma u$ lifts S to $F = (B/x^2,\ A/x,\ \gamma x)$, where $A = \sum_k c_k \gamma^{k-1} u^k + 2$ and $B = \sum_k \frac{k c_k}{2(k+1)} \gamma^{k-1} u^{k+1} + u$, and the divisions are exact precisely when $p(g_0) = -2g_0$, $q(g_0) = -g_0^2$ and $a = -(g_0^2 + g_0 q'(g_0))/(q'(g_0) + 2g_0)$. The first two conditions are linear in c_1, c_2 with system determinant $g_0^4/12$, so any choice of $c_3, \ldots, c_d$ completes to a curve of degree d. Nothing in the derivation is trusted: the divisibilities are verified monomial by monomial, the determinant is expanded and required constant and nonzero, and the two collision witnesses, constructed from a secant line of $\varphi(w) = q(w) - \frac{w}{2}p(w)$ through two rational points, go through the same audit as a published claim. The generator is calibrated before it is believed: at $d = 2$ it must reproduce the announced map polynomial for polynomial (curve $[4, -3]$), and the gate fails if one monomial differs. Gallagher's gauge is implemented alongside: a seed p with $p(0) = 0$, $p(1) = -c$, $\int_0^1 p = 0$, q from $cq' = wp'$, $\kappa = p'(1)/c$, $a = -(1 + \kappa)/(2 + \kappa)$, $F = (\alpha/x^2, \beta/x, x\gamma)$ with $\beta = c + p(w)/\gamma$, $\alpha = u + q(w)/\gamma^2$, and $\det J(F) = bc$; every seed condition is verified and a seed that fails one is refused.

The fiber counter. Given F and a rational target w, a damped multistart Newton hunt in floating point proposes preimages over a ladder of scales; each candidate is then certified by the Krawczyk operator [21] on the *exact* map, with every rational coefficient enclosed by its nearest double widened one unit in the last place each way and strict interior containment required, which gives existence and local uniqueness in an explicit box. Boxes that are provably disjoint hold provably distinct preimages. The output, "at least k preimages of w, each in a certified box", is a lower bound by construction: the hunter reaches what it reaches.

4 Record I: the eleven certificates

Theorem 1 (the published objects, re-decided)**.** *Each of the following holds as decided by the instrument and, independently, by the standard-library verifier, from the pinned sources.*

1. *The announced map $F\colon \mathbb{C}^3 \to \mathbb{C}^3$, with components of degrees 7, 6, 4 and 7, 6, 3 terms, has* $\det J(F) \equiv -2$*, and the three distinct rational points* $(0,\ 0,\ -\frac{1}{4})$, $(1,\ -\frac{3}{2},\ \frac{13}{2})$, $(-1,\ \frac{3}{2},\ \frac{13}{2})$ *all map to* $(-\frac{1}{4},\ 0,\ 0)$*. By Lemma 1 the Jacobian conjecture is false in dimension* 3 *and in every dimension above it; the stabilisation is decided once more through the* 8×8 *symbolic determinant at* $n = 8$*.*

certificate	source	object	n	$\det J$	points	Jacobian monomials	lane
`keller-0`	Alpöge 19 July 2026, via [12]	components of degree 7, 6, 4	3	-2	3	33	transcribed
`keller-1`	the same map, identity-padded	the $n = 3$ map, stabilised once	8	-2	3	38	transcribed
`keller-2`	this repository	curve $[6, -6, 1]$, geometric degree 4	3	-2	2	78	generated
`keller-3`	this repository	curve $[\frac{52}{5}, -\frac{51}{5}, 0, 1]$, geometric degree 5	3	-2	2	141	generated
`keller-4`	this repository	curve $[20, -19, 0, 0, 1]$, geometric degree 6	3	-2	2	222	generated
`keller-5`	Meng–Yang [12]	gradient of Ψ, $\deg \Psi = 14$	5	128	2	339	transcribed
`keller-6`	Gallagher [9]	seed $[2, -3]$, generic fiber 3	3	1	2	33	rebuilt from seed
`keller-7`	Gallagher [9]	seed $[\frac{3}{2}, -\frac{3}{2}, -1]$, generic fiber 4	3	1	2	78	rebuilt from seed
`keller-8`	Gallagher [9]	seed $[\frac{17}{10}, -\frac{27}{10}, 1, -1]$, generic fiber 5	3	1	2	141	rebuilt from seed
`keller-9`	Gallagher [9]	seed $[\frac{9}{5}, -\frac{14}{5}, 0, 1, -1]$, generic fiber 6	3	1	2	222	rebuilt from seed
`keller-10`	Gallagher [9]	seed $[1, 0, -2]$, generic fiber 4	3	-1	2	78	rebuilt from seed

Table 1: The 11 certificates of Record I, every row decided twice at the run that wrote this document. "Points" is the number of distinct rational witnesses sharing one image; "Jacobian monomials" is the size of the symbolic determinant the verifier had to cancel. Geometric and generic fiber degrees in the "object" column are the constructions' nominal values, not decided here (§9).

2. *Meng–Yang's* $\Psi \in \mathbb{Z}[x_1, \dots, x_5]$ *(42 monomials, total degree 14) has* $\det \mathrm{Hess}\, \Psi \equiv 128$*, its gradient has components of degrees 13, 13, 13, 12, 10, and* $\nabla\Psi$ *takes the value* $(0,\ 0,\ -\frac{1}{2},\ 0,\ 0)$ *at the two distinct points* $(1,\ -\frac{3}{2},\ 0,\ 0,\ 0)$ *and* $(-1,\ \frac{3}{2},\ 0,\ 0,\ 0)$*. By Lemma 2,* HC_5 *is false.*

3. *Gallagher's seeds for* $d = 2, 3, 4, 5$ *and the distinct member* $p = w - 2w^3$*,* $b = -1$*, rebuilt under the paper's gauge, give five maps* $\mathbb{C}^3 \to \mathbb{C}^3$ *with* $\det J \equiv bc$*, that is* 1 *and* -1 *(Table 3), each with two distinct rational points sharing one image.*

Proof. Each item is the conjunction of the three facts of Lemma 1 or 2, decided as described in §3; the exact monomial lists, witnesses and images are in the certificate, and the verifier's run is reproduced in the Reproduction section. Across the 11 entries, 1,403 Jacobian monomials had to cancel to the stated constants and did (between 33 and 339 per entry; the Hessian entry is the heaviest at 339), and 24 witness evaluations were compared with their images as fractions. □

Theorem 2 (maps generated here)**.** *For the curves of degree 3, 4, 5 with coefficient vectors* $[c_1, \dots, c_d]$ *listed in Table 2, with* $g_0 = 2$ *and* $b = -1$*, the tangent-sweep recipe yields three polynomial maps* $\mathbb{C}^3 \to \mathbb{C}^3$*, each with* $\det J \equiv -2$ *as a polynomial identity and two distinct rational points sharing one image. By Lemma 1 each is a counterexample to the Jacobian conjecture in dimension three. None of the three is printed in any of the pinned sources.*

Proof. The recipe's divisibility conditions are verified monomial by monomial and the three facts are decided exactly as for a published claim; the certificate records the curve, the twist coefficient a, the secant pair and the witnesses, and the verifier re-derives the determinant from the monomial lists alone. □

A remark on what "generated here" means. A new curve through a published mechanism is a new instance, not a new mechanism; the construction is Speyer's as written up by Gao, and this repository claims none of it. The three maps have not been checked for coordinate equivalence against the members of the published family (§8). What is certified is what the certificate says: these polynomials, this determinant, these two points.

Proposition 1 (the doubling identity, recomputed)**.** *Let* F *be the announced map and* $\varphi = y \cdot F = \sum_{i=1}^{3} y_i F_i$ *in six variables (16 monomials). Then* $\det \mathrm{Hess}\, \varphi \equiv -4 = -(\det J(F))^2$*.*

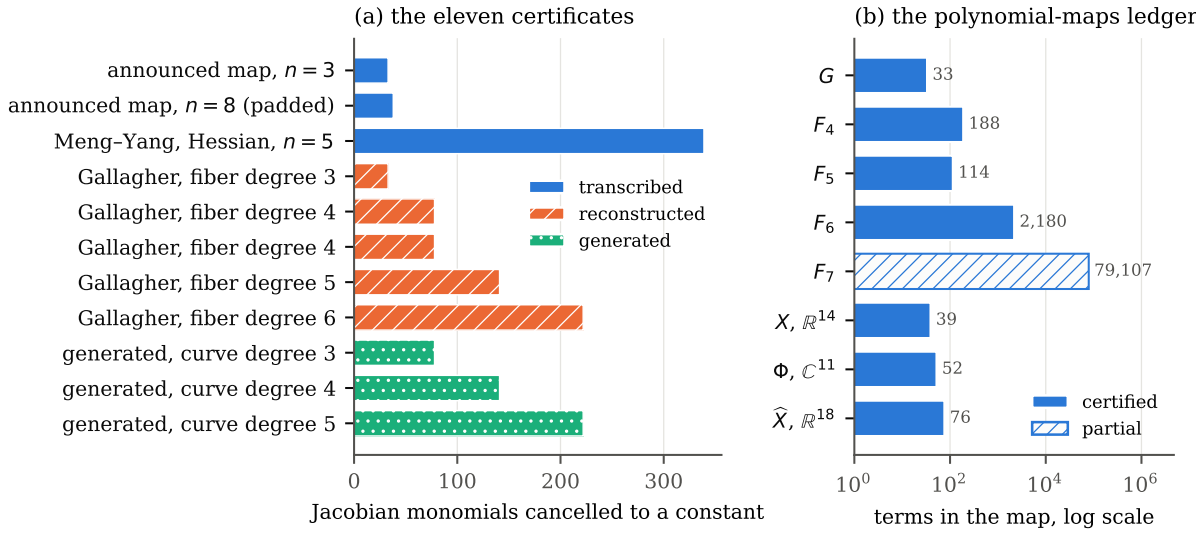


Figure 1: The exact-arithmetic load of each decision, drawn from the records. (a) For each certificate of Record I, the number of monomials in the symbolic Jacobian that the independent verifier cancelled to a constant, grouped by lane. (b) For each object of Record II, the number of terms in the map; F_7, hatched, is the one row decided in part.

certificate	curve $[c_1, \dots, c_d]$	a	nominal geom. degree	component degrees	Jacobian monomials	the two witnesses
`keller-2`	$[6, -6, 1]$	-4	4	12, 11, 4	78	$(-\frac{2}{3}, \frac{5}{2}, \frac{183}{8})$, $(\frac{2}{11}, -\frac{13}{2}, \frac{297}{8})$
`keller-3`	$[\frac{52}{5}, -\frac{51}{5}, 0, 1]$	$-\frac{9}{7}$	5	17, 16, 4	141	$(-\frac{5}{11}, \frac{16}{5}, \frac{25707}{875})$, $(\frac{5}{41}, -\frac{46}{5}, -\frac{279907}{875})$
`keller-4`	$[20, -19, 0, 0, 1]$	$-\frac{13}{7}$	6	22, 21, 4	222	$(-\frac{6}{25}, \frac{31}{6}, \frac{222325}{1512})$, $(\frac{6}{101}, -\frac{107}{6}, -\frac{5512277}{1512})$

Table 2: The 3 generated maps: new curves through the published mechanism, with $g_0 = 2$, $b = -1$ and $\det J \equiv -2$. The nominal geometric degree is $d + 1$ by the construction; what is decided is $\det J$ and the two witnesses, and for the first map a blind count of at least three preimages (Table 4, whose targets are these maps' images).

Proof. The 6×6 Hessian (126 monomials in total) is expanded symbolically at every build and at the run that wrote this document; the constant is -4. This is the six-variable Hessian counterexample Meng–Yang record as a warm-up, and the identity is what ties the two transcribed entries of Record I to each other: the five-variable Ψ descends from φ by a Schur complement, and if the transcription of either object were wrong the recomputed constant would not be $-(-2)^2$. Both entries pin the same file, so this is an arithmetic independence and not a second source. □

5 Record I, continued: the collisions found blind

Everything in §4 decides witnesses somebody supplied. This section supplies none. For each three-dimensional map of Record I and a rational target, the fiber counter of §3 hunts preimages with 400 Newton starts over a ladder of scales and certifies every candidate it reaches in a Krawczyk box on the exact map.

Theorem 3 (blind fiber counts)**.** *For each cell of Table 4 with verdict "certified", the map has*

certificate	seed $[c_1, \ldots, c_d]$	a	b	$\det J$	nominal fiber degree	the two witnesses
`keller-6`	$[2, -3]$	$-\frac{3}{2}$	1	1	3	$(-\frac{1}{3},\ 4,\ -54)$, $(\frac{1}{4},\ -2,\ 36)$
`keller-7`	$[\frac{3}{2}, -\frac{3}{2}, -1]$	$-\frac{7}{5}$	1	1	4	$(2,\ \frac{1}{2},\ \frac{9}{40})$, $(\frac{2}{3},\ -\frac{5}{2},\ -\frac{33}{8})$
`keller-8`	$[\frac{17}{10}, -\frac{27}{10}, 1, -1]$	$-\frac{37}{27}$	1	1	5	$(-10,\ \frac{11}{10},\ -\frac{4367}{27000})$, $(\frac{10}{53},\ -\frac{63}{10},\ \frac{225091}{3000})$
`keller-9`	$[\frac{9}{5}, -\frac{14}{5}, 0, 1, -1]$	$-\frac{19}{14}$	1	1	6	$(\frac{15}{4},\ \frac{11}{15},\ \frac{5038}{23625})$, $(\frac{15}{28},\ -\frac{43}{15},\ -\frac{14318}{3375})$
`keller-10`	$[1, 0, -2]$	$-\frac{4}{3}$	-1	-1	4	$(1,\ 0,\ 0)$, $(-1,\ 0,\ 2)$

Table 3: Gallagher's family rebuilt from its seeds under the paper's gauge ($c = 1$ throughout; $\det J = bc$). The seeds for $d = 2, \ldots, 5$ are the paper's $p_d = 2w - 3w^2 + w(1-w)(w^{d-2} - 6/(d(d+1)))$, and the last row is its distinct member $p = w - 2w^3$ with $b = -1$, whose twist coefficient $a = -\frac{4}{3}$ matches the paper.

cell	map	target	target from	witnesses constructed	certified blind	verdict
`alpoge`	the announced map	$(-\frac{1}{4},\ 0,\ 0)$	own collision image	3	3	≥ 3 preimages, certified
`alpoge-own-target`	the announced map	$(0,\ 1,\ 0)$	chosen here	2	3	≥ 3 preimages, certified
`sweep-d3`	generated, curve degree 3	$(-\frac{13}{8},\ -2,\ 1)$	own collision image	2	3	≥ 3 preimages, certified
`sweep-d4`	generated, curve degree 4	$(-\frac{13}{5},\ -\frac{16}{5},\ 1)$	own collision image	2	2	≥ 2 preimages, certified
`sweep-d5`	generated, curve degree 5	$(-\frac{61}{12},\ -\frac{19}{3},\ 1)$	own collision image	2	0	no certified box
`gallagher-d2`	Gallagher seed $d = 2$	$(-4,\ -4,\ 1)$	own collision image	2	3	≥ 3 preimages, certified
`gallagher-d3`	Gallagher seed $d = 3$	$(-\frac{1}{2},\ -\frac{1}{2},\ 1)$	own collision image	2	3	≥ 3 preimages, certified
`gallagher-d4`	Gallagher seed $d = 4$	$(-\frac{11}{10},\ -\frac{11}{10},\ 1)$	own collision image	2	1	one box: nothing proved
`gallagher-distinct`	Gallagher distinct member	$(0,\ 0,\ 1)$	own collision image	2	2	≥ 2 preimages, certified

Table 4: The fiber family, re-run at the run that wrote this document: 7 of 9 cells certify non-injectivity blind. A count is a lower bound; a cell that returns one box (REJECT) or none (REFUSED) proves nothing and says so.

at least the stated number of preimages of the stated target, each in an explicit box in which it is the unique preimage, the boxes pairwise disjoint. In particular the announced map has at least three preimages of $(-\frac{1}{4},\ 0,\ 0)$ *and at least three preimages of* $(0,\ 1,\ 0)$*, so it is not injective, with no published witness consumed.*

Proof. Krawczyk's criterion with strict interior containment on interval-enclosed exact coefficients [21]; disjoint boxes hold distinct points. For the first cell each of the three published witnesses lies inside one of the three certified boxes, which is the check that the rediscovery is of the same points and not of three others; the hunt never read them. The target $(0,\ 1,\ 0)$ of the second cell was chosen here by a fixed enumeration of plain rational points, $(0,0,0), (1,1,1), (1,0,0), (0,1,0), \ldots$, the first whose blind count certifies at least two; it is nobody's published image and nobody's constructed collision. □

Three facts in the table deserve attention. In 3 cells (`sweep-d3`, `gallagher-d2`, `gallagher-d3`) the hunter certifies more preimages than the two witnesses its own construction wrote down, consistent with the nominal geometric degrees of those maps but proving only the lower bound. In two cells (`gallagher-d4` and `sweep-d5`) the hunt does not certify non-injectivity at all: the float layer did not reach enough roots, and the certificate reports one box or none rather than rounding up. Those maps are still certified non-injective by Theorem 1 and Theorem 2; the counter simply did not re-find it unaided, and a lower bound that is allowed to be weak is the only kind that can be trusted when it is strong. Finally, the control that makes the counts mean something: a preimage counter that manufactured non-injectivity would find extra roots everywhere, so it is run against the triangular automorphism $(x + y^2,\ y + z^3,\ z)$, injective by construction with $\det J = 1$; over 400 starts on the same ladder the dedup by certified box disjointness returns exactly one preimage.

6 Record II: Gao's five maps

Gao's paper [13] (arXiv 2608.00222v1) prints G, F_4 and F_5 in full, and F_6 and F_7 through their construction: four potentials, a direction field, a weighted scaling, a stage map and a monomial substitution, with the first component printed and the term counts of the others stated. The paper reports that "All polynomial identities asserted below … were verified by exact rational arithmetic; the verification scripts are available from the author." (p. 2); pp. 24 and 28 promise ancillary files with F6 and F7, which arXiv does not carry (as recorded in the corpus metadata when the sources were pinned). The verdicts on F_6 and F_7 are therefore on the map the printed construction defines: the decider rebuilds each from the printed recipe, checking every division exact, and checks the rebuild against every piece the paper does print (the four printed polynomials $X_1, \dots, X_4$ and the first component for F_6; the first component for F_7). The non-injectivity witnesses are found by this repository from each paper's fiber structure, by a proposer the decider does not trust: it only evaluates them.

Theorem 4 (Gao's maps, decided)**.** *Decided in exact rational arithmetic from the pinned TeX, at the ledger of 2026-10-05:*

1. $G\colon \mathbb{C}^3 \to \mathbb{C}^3$ *(33 terms) has* $\det J(G) \equiv 2$ *and two distinct rational points with image* $(1, \frac{5}{4}, 0)$.
2. $F_4\colon \mathbb{C}^4 \to \mathbb{C}^4$ *(188 terms) has* $\det J(F_4) \equiv -\frac{44}{9}$ *and 3 distinct rational points with image* $(1, -\frac{1289}{270}, -\frac{113}{90}, \frac{1453}{90})$.
3. $F_5\colon \mathbb{C}^4 \to \mathbb{C}^4$ *(114 terms) has* $\det J(F_5) \equiv \frac{160}{29}$ *and two distinct rational points with image* $(1, -\frac{89}{116}, -\frac{605}{116}, -\frac{109}{58})$.
4. $F_6\colon \mathbb{C}^5 \to \mathbb{C}^5$*, rebuilt from the printed construction with component degrees 7, 38, 40, 42, 44 and 6, 342, 421, 507, 904 terms (2,180 in all), has* $\det J(F_6) \equiv -290$ *by direct expansion of the* 5×5 *determinant, and two distinct rational points with image* $(1, 1, 2, -508, -660)$.

By Lemma 1, each is a counterexample to the Jacobian conjecture in its dimension. The component degrees agree with the printed ones in every case.

Proposition 2 (F_7, decided in part)**.** *The map* $F_7\colon \mathbb{C}^5 \to \mathbb{C}^5$ *that the printed construction defines has component degrees 7, 86, 89, 92, 95 and 7, 14,445, 17,727, 21,410, 25,518 terms, reproduces the printed first component, and sends two distinct rational points to one image, so it is not injective. The determinants of its factors are as the paper states:* $\det J(S) = \gamma$ *for the sweep,* 119377 *for the affine stage, and* $\det J_{(w_2,w_3)}(H_3, H_4) = 1$ *for the transverse data. The constancy of* $\det J(F_7)$ *itself is not decided here.*

Why it is partial. The whole 5×5 determinant has rows of between 14,445 and 25,518 terms, and its direct expansion is out of reach of the pure-Python decider in the time the ledger allows; the constant rests on the paper's chain factorisation $x^{10} \cdot 119377 \cdot \gamma^8 \cdot \gamma^2 \cdot (\gamma x)^{-10}$, whose factors are decided and whose assembly is the paper's lemma. The ledger records the check with a null result, the verdict PARTIAL, and the row names what would close it: a formalised chain rule for the factorisation, or a determinant routine that expands rows of that size. □

The paper's generic fiber sizes (Table 5, and 3 for the announced map) are not decided in either record. Their proofs in [13] run through univariate elimination, squarefreeness and gcd certificates at witness targets; the instruments here decide lower bounds only (Theorem 3), and the exact counts are listed in §9 as the first thing a next version should close.

map	n	component degrees	terms	$\det J$	how decided	points	generic fiber (paper)	verdict
G	3	4, 11, 12	33	2	identity	2	4	CERTIFIED
F_4	4	4, 11, 12, 21	188	$-\frac{44}{9}$	identity	3	5	CERTIFIED
F_5	4	3, 12, 14, 16	114	$\frac{160}{29}$	identity	2	10	CERTIFIED
F_6	5	7, 38, 40, 42, 44	2,180	-290	identity	2	6	CERTIFIED
F_7	5	7, 86, 89, 92, 95	79,107	119377	factors only	2	12	PARTIAL

Table 5: Gao's five maps. "Terms" counts all components; "points" is the number of distinct rational points sharing one image; "generic fiber" is the paper's theorem statement, read from the pinned TeX and not decided here. The whole ledger re-derives in about 90 seconds.

7 Record II, continued: the weak Markus–Yamabe fields

The paper of Castañeda, Honorato and Valenzuela-Henríquez [14] (arXiv 2608.05392v1) prints three objects: the field X on $\mathbb{R}^{14}$ obtained from the announced map by the chain realisation, the cubic map Φ on eleven variables that is the map's cubic stabilisation in the sense of Bass–Connell–Wright, and the cubic field $\widehat{X}$ on $\mathbb{R}^{18}$ obtained from Φ by a rank-reduced dehomogenisation. The chain dimension is computed from the announced map's component degrees as recorded in Record I: $\sum_i \max(d_i, 2) - n = 17 - 3 = 14$, which is the dimension of the field the ledger decides.

Theorem 5 (the fields, decided)**.** *Decided in exact rational arithmetic from the pinned TeX, at the same ledger:*

1. $X\colon \mathbb{R}^{14} \to \mathbb{R}^{14}$*, of degree 7 with 39 terms, satisfies* $(JX + I)^{14} = 0$ *identically (and no lower power vanishes), and vanishes at the 3 distinct rational points the paper prints. By Lemma 3 the weak Markus–Yamabe conjecture is false in dimension* 14 *and in every dimension above it.*

2. $\widehat{X}\colon \mathbb{R}^{18} \to \mathbb{R}^{18}$*, of degree 3 with 76 terms, satisfies* $(J\widehat{X} + I)^{17} = 0$ *identically (and no lower power vanishes), and vanishes at 3 distinct rational points. The weak conjecture is therefore also false in dimension* 18 *by a field of degree three.*

3. $\Phi\colon \mathbb{C}^{11} \to \mathbb{C}^{11}$*, cubic with 52 terms, has* $\det J(\Phi) \equiv -2$ *by a direct* 11×11 *expansion, and 3 distinct points share one image; it is a cubic counterexample to the Jacobian conjecture in dimension* 11*.*

Proof. The matrix powers are multiplied out as matrices of exact polynomials; the zeros and images are evaluated as fractions. For $\widehat{X}$ the pinned appendix checks the unipotency "at a sample rational point" with a computer-algebra system; here the vanishing of the seventeenth power is an identity, which is what Lemma 3 needs at every point. For Φ the paper verifies the determinant through a 3×3 Schur complement; the direct expansion decides the same constant. □

A note on provenance, recorded in the corpus metadata rather than decided: Phi (Prop 6.2) equals, all 11 components, the map of gist.github.com/Spacerat/08b4a43f6b6ca57178efabc220170ce8, which says: "The explicit 11-variable construction, simplification, and verification code in this file were generated by ChatGPT (OpenAI)." The appendix checks the 18-dimensional unipotency "at a sample rational point". The certificate is indifferent to who wrote the map; the fact is recorded because the decision grammar of this paper exists for exactly such objects.

object	N	degree	terms	identity decided	zeros / points	verdict
X on $\mathbb{R}^{14}$	14	7	39	$(JX+I)^{14}=0$	3 zeros	CERTIFIED
Φ on $\mathbb{C}^{11}$	11	3	52	$\det J\Phi=-2$	3 points, one image	CERTIFIED
$\widehat{X}$ on $\mathbb{R}^{18}$	18	3	76	$(J\widehat{X}+I)^{17}=0$	3 zeros	CERTIFIED

Table 6: The three objects of [14], as decided. For Φ the identity is a constant determinant and the points are three with one image; for the fields it is the nilpotency of $JX+I$ as a matrix of polynomials, with the index that first vanishes.

8 The moduli question, open

The generator of §3 is a parametrised family. At curve degree d the coefficients $c_3,\dots,c_d$ are free, c_1, c_2 are determined by the two divisibility conditions, and g_0 and b are free nonzero constants; the three shipped instances take $g_0=2$, $b=-1$, $c_d=1$ and the remaining free coefficients zero. Gallagher's gauge parametrises the same mechanism differently, and the paper's family realises every generic fiber degree $d\geq 3$ in dimension three. Composition with polynomial automorphisms on either side identifies many of these maps, and, as [13] notes, geometric degree is invariant under such composition; beyond that invariant nothing is decided here, and no equivalence-testing instrument exists in the repository.

Question. For each geometric degree $k\geq 3$, how many coordinate-inequivalent Keller counterexamples $\mathbb{C}^3\to\mathbb{C}^3$ does the tangent-sweep family contain, and is every dimension-three counterexample equivalent to one of them? The three generated maps of Theorem 2 have nominal geometric degrees 4, 5, 6; whether the first is equivalent to Gallagher's $d=3$ member or to the distinct member (both of nominal degree 4), or the second and third to the $d=4$ and $d=5$ members, is not known to us. The record labels every generated row "not coordinate-equivalence-checked" for this reason.

9 What is not claimed

The certificates are narrow on purpose; the value of a narrow claim is that it is either true or refuted. Stated plainly:

- *Not that the external papers stand.* Each is cited as a claim and its bytes are pinned. Whether it is correct as a paper, survives refereeing, or has been contested since the pinned version is not decided here. Every verdict concerns the explicit object printed in the record, and PARTIAL on F_7 measures this instrument's reach, not the paper.
- *No priority, in either direction.* Sources are recorded as transcribed, with the dates they carry. The generated maps are new instances of a published mechanism.
- *Nothing about the plane.* Every map here has $n\geq 3$; the two-dimensional case is untouched, as is the Hessian conjecture in four variables.
- *One stabilisation, not six theorems.* Identity padding carries a counterexample to every higher dimension unchanged; the record carries one padded row, at $n=8$, whose certificate text says it is padding.

- *Lower bounds only on fibers.* The blind counts are "at least k". The generic fiber sizes of Gao's theorems (4, 5, 10, 6, 12 and 3 for the announced map) and the nominal geometric degrees of the generated and rebuilt maps are not decided; they would need squarefreeness and gcd certificates at witness targets together with certified boxes, which are not built.
- *F_7's whole determinant.* Only its factors are decided (Proposition 2).
- *The chain cannot be shortened.* The Markus–Yamabe paper states that moving any homogeneous part one level up destroys the nilpotency and that this was verified in exact arithmetic; it prints no certificate, and the finite family of variants is not decided here.
- *Nothing beyond the identities for the dynamics.* Hyperbolic sinks, almost global stability and the discrete counterpart in [14] are consequences the paper draws from the spectrum; only the spectrum, as the nilpotency identity, and the zeros are decided.
- *No proof by a language model.* Models are in no decision path. The instrument, the generator, the fiber counter, the batteries and the verifier are deterministic code, and every verdict is a finite exact computation a reader can re-run.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. Record I: `certs/keller-certificate.json` (the 11 certificates, every polynomial as explicit monomials with exact rational coefficients), written by `tools/export-keller-certificate.js` from `families/keller-audit.js`; `python3 tools/verify_keller.py certs/keller-certificate.json --sources corpus/sources` re-decides all 11 in a fraction of a second with the standard library only and must refute a forged coefficient before exiting green; `node instruments/keller/battery.js` runs the instrument's gate (32 checks, 9 red controls, the generator calibration and the blind rediscovery); `node tools/build-report-keller.js` rebuilds the public page `https://carlostoledo.co/reports/keller.html`, re-running every gate, the fiber family (`families/keller-fibers.js`, about ten seconds) and the doubling identity. The instrument is `instruments/keller/` (`keller.js` the audit, `sweep.js` the generator and Gallagher's gauge, `fibers.js` the Krawczyk fiber counter). Sources pinned in `corpus/sources/`: `gallagher2026.pdf` and `mengyang2026.pdf` (hashes in `PINS.json` and in the certificate). Record II: `certs/polymaps-ledger.json`, written by `tools/run-polymaps-le` from `instruments/polymaps/decide.py` over `corpus/polymaps/` (the two TeX sources under CC BY 4.0, `maps.json` extracted by `extract.py`, the witnesses in `witnesses.json`); `python3 tools/run-polymaps-ledger.py --check` re-derives it in about 90 seconds and refuses on any difference; `python3 instruments/polymaps/battery.py` runs its gate; the page is `https://carlostoledo.co/reports/polymaps.html`. This document: `node tools/paper-numbers/jacobian-counter` writes every macro from the two records and the live runs (git 16849460e608), `tools/paper-numbers/jacobian-c` draws Figure 1, and `node tools/build-paper-tex.js jacobian-counterexamples` compiles. The repository is archived under concept DOI 10.5281/zenodo.22225860.

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